Nonlinear Optimization Project: Planning Production and Transmission of Electric Power

Group 46
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Abstract

This report presents a simplified model for planning the production and transmission of electric power in a small-scale network. The model includes only plannable power plants, and aims to minimize the cost of active power production while satisfying consumer demand. Reactive power flows and voltage constraints are also considered.

Contents

1 Introduction

In this project, we develop a simplified model for planning production and transmission of electric power. Only production from plannable power plants (hydro, thermal, nuclear) is considered, whereas solar and wind power production is assumed to be already allocated. Table 1 lists the largest 14 plannable power plants in Sweden. Figure 1 illustrates the simplified transmission grid used in this project.

Table 1: Largest plannable power plants in Sweden.

Name	Capacity [MW]	Type
Forsmarks	3271	nuclear
Ringhals	2190	nuclear
Oskarshamn	1450	nuclear
Harsprångets	977	hydro
Karlshamnsverket	662	oil
Stornorrfors	599	hydro
Värtaverket	579	biomass; oil; coal
Stenungsunds	520	oil
Letsi	486	hydro
Porjus	481	hydro
Messaure	446	hydro
Ligga	327	hydro
Vietas	306	hydro
Trängslet	300	hydro

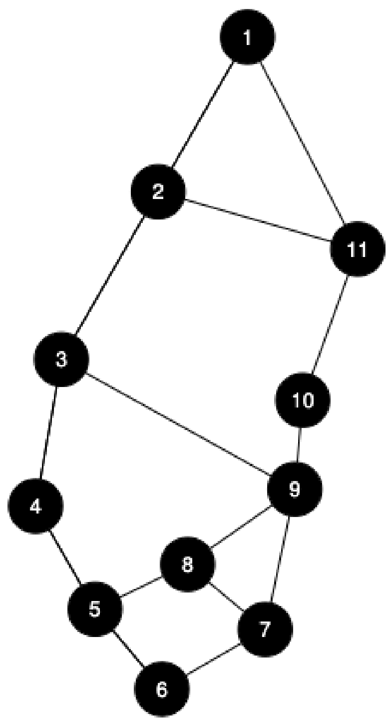

Figure 1: Simplified map of power plants and the transmission grid.

2 Simplified Transmission Grid Model

For each node k in the network, we define the voltage amplitude v_k and voltage phase angle θ_k. The active and reactive power flows between nodes k and ℓ are given by:

$$p_{k\ell} = v_k^2 g_{k\ell} - v_k v_\ell g_{k\ell} \cos(\theta_k - \theta_\ell) - v_k v_\ell b_{k\ell} \sin(\theta_k - \theta_\ell), \quad (1)$$
$$q_{k\ell} = -v_k^2 b_{k\ell} + v_k v_\ell b_{k\ell} \cos(\theta_k - \theta_\ell) - v_k v_\ell g_{k\ell} \sin(\theta_k - \theta_\ell), \quad (2)$$

where $g_{k\ell}$ and $b_{k\ell}$ are parameters describing the transmission line between nodes k and ℓ. Voltages are in pu (per unit) and angles in radians.

2.1 Generator and Consumer Data

Table 2 lists the generators and their parameters.

Table 2: Generator parameters

Generator	Node	Max Capacity [pu]	Cost [SEK/pu]
G1	2	0.02	175
G2	2	0.15	100
G3	2	0.08	150
G4	3	0.07	150
G5	4	0.04	300
G6	5	0.17	350
G7	7	0.17	400
G8	9	0.26	300
G9	9	0.05	200

Table 3 lists the consumers and their active power demands.

Table 3: Consumer parameters

Consumer	Node	Demand [pu]
C1	1	0.10
C2	4	0.19
C3	6	0.11
C4	8	0.09
C5	9	0.21
C6	10	0.05
C7	11	0.04

3 Mathematical Formulation

3.1 Sets

- $\mathcal{G} = \{G_1, G_2, \ldots, G_9\}$: set of generators

- $\mathcal{C} = \{C_1, C_2, \ldots, C_7\}$: set of consumers
- $\mathcal{N} = \{1, 2, \ldots, N_n\}$: set of all nodes in the network
- $\mathcal{L} = \{(k, \ell) \,|\, k \text{ and } \ell \text{ are connected}\}$: set of transmission lines
- $\mathcal{G}_k \subset \mathcal{G}$: set of generators located at node k
- $\mathcal{C}_k \subset \mathcal{C}$: set of consumers located at node k

3.2 Decision Variables

- P_i: active power generated by generator $i \in \mathcal{G}$
- Q_i: reactive power generated by generator $i \in \mathcal{G}$
- v_k: voltage magnitude at node $k \in \mathcal{N}$
- θ_k: voltage phase angle at node $k \in \mathcal{N}$
- $p_{k\ell}$, $q_{k\ell}$: active and reactive power flow from node k to node ℓ, $(k, \ell) \in \mathcal{L}$

3.3 Parameters

- c_i: cost per unit of active power produced by generator $i \in \mathcal{G}$
- P_j^{load}: active power demand of consumer $j \in \mathcal{C}$
- Q_j^{load}: reactive power demand of consumer $j \in \mathcal{C}$
- $P_i^{\max}$: maximum active power capacity of generator $i \in \mathcal{G}$
- $g_{k\ell}, b_{k\ell}$: conductance and susceptance of the line between nodes k and ℓ

3.4 Optimization Problem

Objective: Minimize total active power cost

$$\min_{P_i} \quad \sum_{i \in \mathcal{G}} c_i P_i \tag{3}$$

Subject to:

$$\sum_{i \in \mathcal{G}_k} P_i - \sum_{j \in \mathcal{C}_k} P_j^{\text{load}} = \sum_{\ell \in \text{neighbors}(k)} p_{k\ell}, \qquad \forall k \in \mathcal{N} \tag{4}$$

$$\sum_{i \in \mathcal{G}_k} Q_i = \sum_{\ell \in \text{neighbors}(k)} q_{k\ell}, \qquad \forall k \in \mathcal{N} \tag{5}$$

$$0 \leq P_i \leq P_i^{\max}, \qquad \forall i \in \mathcal{G} \tag{6}$$

$$-0.03 P_i^{\max} \leq Q_i \leq 0.03 P_i^{\max}, \qquad \forall i \in \mathcal{G} \tag{7}$$

$$0.98 \leq v_k \leq 1.02, \qquad \forall k \in \mathcal{N} \tag{8}$$

$$-\pi \leq \theta_k \leq \pi, \qquad \forall k \in \mathcal{N} \tag{9}$$

Equations (3)–(9) define the nonlinear optimization model for minimizing the total cost of active power generation while ensuring network feasibility.

- **Objective (3):** Minimizes the total production cost of active power, where c_i represents the unit generation cost of generator i, and P_i is its active power output.
- **Constraint (4):** Enforces the *active power balance* at each node k. The total generated active power minus local consumption must equal the net power flow to neighboring nodes.
- **Constraint (5):** Represents the *reactive power balance* at each node. The reactive power generated by all generators connected to node k must equal the total reactive power flows to adjacent nodes.
- **Constraints (6)–(7):** Impose the *generation limits*. Each generator i can only produce up to its maximum active power $P_i^{\max}$, and its reactive power Q_i must remain within $\pm 3\%$ of that maximum.
- **Constraint (8):** Ensures that the *voltage magnitude* v_k at each node stays within operational limits (0.98–1.02 pu), as required for stable grid operation.
- **Constraint (9):** Restricts the *voltage angle* θ_k at each node to the interval $[-\pi, \pi]$, ensuring physically meaningful phase differences.

3.5 Results and analysis

Summary of requested outputs. The following presents the numerical answers requested by the assignment: the total cost of production, each generator's active and reactive production, node voltages (magnitudes and phase angles), and the active and reactive power flows on the transmission lines.

3.5.1 Total cost of production

Using the generator outputs obtained from the numerical solution and the unit costs from Table 3, the total cost of active power production equals

$$\boxed{\text{Total cost } = 183.244 \text{ SEK}}$$

(rounded to three decimals). The per-generator cost contributions are shown in the table below.

3.5.2 Generator outputs (active and reactive power)

Table 4: Generator outputs. P rounded to 3 decimals, Q to 5 decimals.

Generator	Node	P [pu]	Q [pu]
G1	2	0.020	-0.00006
G2	2	0.150	-0.00187
G3	2	0.080	-0.00072
G4	3	0.070	0.00180
G5	4	0.040	0.00119
G6	5	0.121	0.00061
G7	7	0.000	0.00096
G8	9	0.260	0.00164
G9	9	0.050	-0.00011

Per-generator cost contributions (SEK):

G1: 3.500
G2: 15.000
G3: 12.000
G4: 10.500
G5: 12.000
G6: 42.244
G7: 0.000
G8: 78.000
G9: 10.000

(rounded to three decimals; sum = 183.244 SEK as above).

3.5.3 Node voltages and angles

Voltages are shown as magnitudes v_k (rounded to 3 decimals) and angles θ_k (radians, rounded to 5 decimals):

Table 5: Node voltages and angles

Node	v_k [pu]	θ_k [rad]
1	1.019	0.00000
2	1.020	0.00454
3	1.019	0.00065
4	1.018	-0.00815
5	1.018	-0.00463
6	1.017	-0.00922
7	1.018	-0.00573
8	1.017	-0.00678
9	1.019	-0.00076
10	1.019	-0.00139
11	1.019	0.00005

3.5.4 Active and reactive power flows on edges

Below are the active power flow matrix $p_{k\ell}$ (rounded to 3 decimals) and the reactive power flow matrix $q_{k\ell}$ (rounded to 5 decimals). Rows are source node k, columns are destination node ℓ. (Entries that are zero have been written as 0.000 or 0.00000 for clarity.)

Active power flow matrix $p_{k\ell}$ (3 decimals):

$$p_{k\ell} = \begin{bmatrix}
0.000 & -0.098 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & -0.002 \\
0.098 & 0.000 & 0.069 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.082 \\
0.000 & -0.069 & 0.000 & 0.110 & 0.000 & 0.000 & 0.000 & 0.000 & 0.029 & 0.000 & 0.000 \\
0.000 & 0.000 & -0.110 & 0.000 & -0.040 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\
0.000 & 0.000 & 0.000 & 0.040 & 0.000 & 0.059 & 0.000 & 0.022 & 0.000 & 0.000 & 0.000 \\
0.000 & 0.000 & 0.000 & 0.000 & -0.059 & 0.000 & -0.051 & 0.000 & 0.000 & 0.000 & 0.000 \\
0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.051 & 0.000 & 0.011 & -0.061 & 0.000 & 0.000 \\
0.000 & 0.000 & 0.000 & 0.000 & -0.022 & 0.000 & -0.011 & 0.000 & -0.058 & 0.000 & 0.000 \\
0.000 & 0.000 & -0.029 & 0.000 & 0.000 & 0.000 & 0.061 & 0.058 & 0.000 & 0.009 & 0.000 \\
0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & -0.009 & 0.000 & -0.041 \\
0.002 & -0.082 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.041 & 0.000
\end{bmatrix}$$

Reactive power flow matrix $q_{k\ell}$ (5 decimals):

$$q_{k\ell} = \begin{bmatrix}
0.00000 & 0.00228 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & -0.00228 \\
-0.00184 & 0.00000 & -0.00092 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00010 \\
0.00000 & 0.00118 & 0.00000 & 0.00296 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & -0.00234 & 0.00000 & 0.00000 \\
0.00000 & 0.00000 & -0.00199 & 0.00000 & 0.00318 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\
0.00000 & 0.00000 & 0.00000 & -0.00304 & 0.00000 & -0.00154 & 0.00000 & 0.00519 & 0.00000 & 0.00000 & 0.00000 \\
0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00181 & 0.00000 & -0.00181 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\
0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00199 & 0.00000 & 0.00524 & -0.00626 & 0.00000 & 0.00000 \\
0.00000 & 0.00000 & 0.00000 & 0.00000 & -0.00514 & 0.00000 & -0.00522 & 0.00000 & 0.01036 & 0.00000 & 0.00000 \\
0.00000 & 0.00000 & 0.00238 & 0.00000 & 0.00000 & 0.00000 & 0.00658 & -0.01003 & 0.00000 & 0.00261 & 0.00000 \\
0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & -0.00260 & 0.00000 & 0.00260 \\
0.00228 & 0.00027 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & -0.00254 & 0.00000
\end{bmatrix}$$

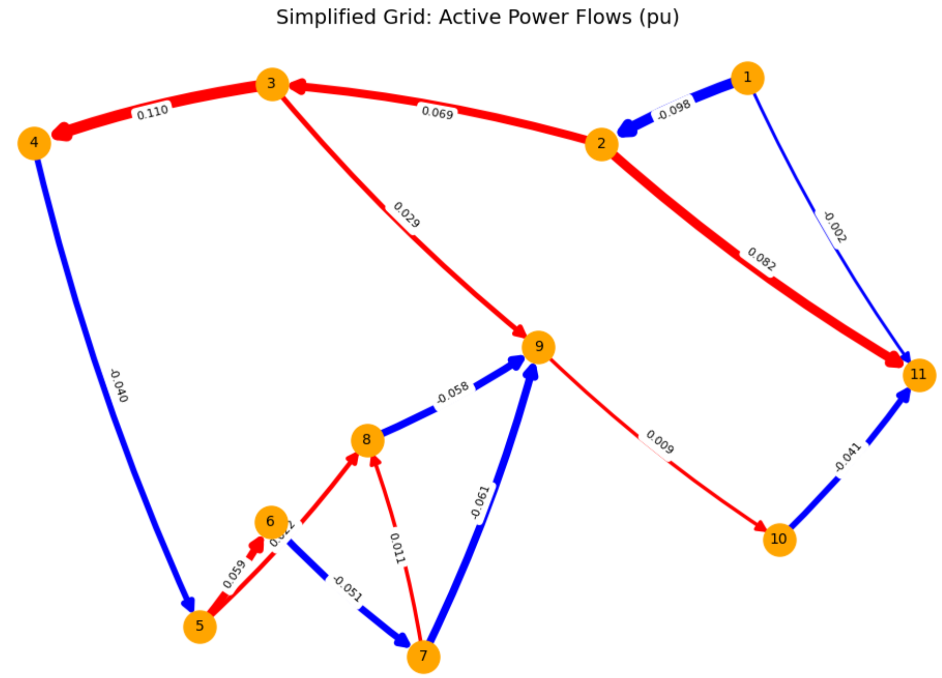

Figure 2: Simplified grid showing active power flows between nodes (in pu). Red arrows indicate export (positive flow), blue arrows indicate import (negative flow), and arrow thickness is proportional to the flow magnitude.

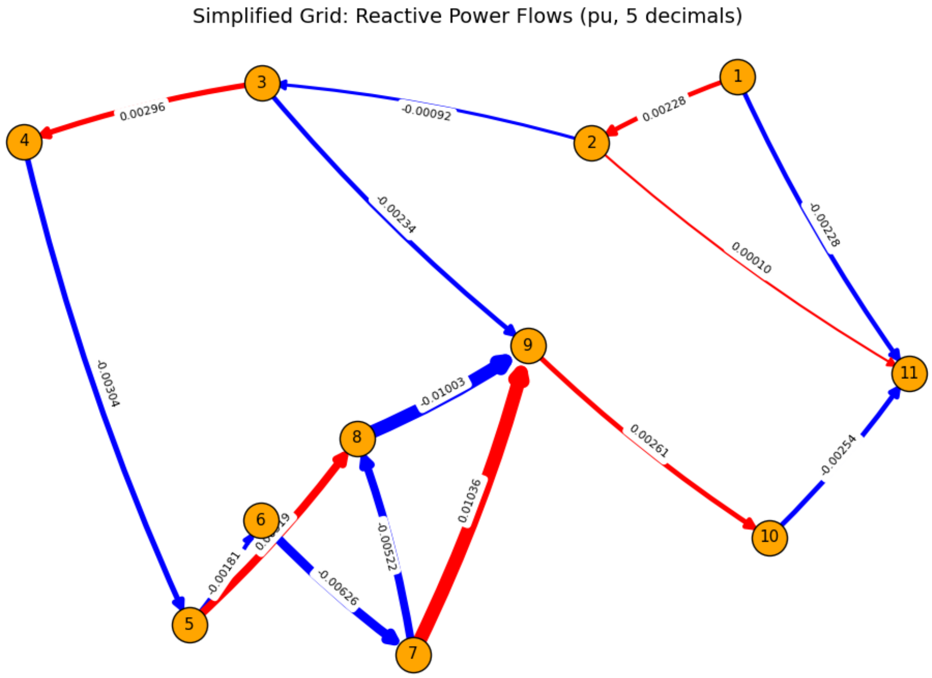

Figure 3: Simplified grid showing reactive power flows between nodes (in pu). Red arrows indicate export (positive flow), blue arrows indicate import (negative flow), and arrow thickness is proportional to the flow magnitude.

3.5.5 Discussion and interpretation of results

The total cost of power production is **183.244** SEK. All generators are producing at or near their maximum capacity except the two most expensive units, G6 and G7, which produce only what is necessary to meet the system demand. This is consistent with an economic dispatch that prioritizes low-cost generation first.

Generator active and reactive power outputs are shown in Table 4. Reactive power contributions are small; positive Q_i indicates generation and negative Q_i indicates absorption, which helps maintain voltage stability.

Node voltages remain within safe limits (0.98–1.02 pu) and phase angles are small (approximately -0.009 to 0.005 rad), indicating a stable, lightly loaded system.

The active and reactive power flow matrices, $p_{k\ell}$ and $q_{k\ell}$, indicate the direction and magnitude of power transfer between nodes. Nonzero entries correspond to actual network edges.

In summary, the system meets all demand while minimizing cost: low-cost units are fully utilized, high-cost units supply the remaining power, voltages are within bounds and no constraints are violated.

4 Summary and Conclusion

The optimization problem for minimizing total active power cost is nonconvex due to the nonlinear AC power flow equations, which involve sine and cosine functions of the voltage angles. While the objective function and generator limits are linear, the nonlinear flows make the feasible set nonconvex. As a result, the solution obtained from the numerical solver is feasible and satisfies all constraints, but can only be guaranteed to be *locally optimal*, not globally optimal.

Despite this, the solution provides meaningful insights for planning: low-cost generators are fully utilized, higher-cost generators operate as necessary to meet demand, voltage magnitudes remain within safe limits, and reactive power flows are small. This allocation aligns with economic dispatch principles and represents an efficient, practically implementable operating point for the power network.

A Code

```
using JuMP
using Ipopt

# === Sets ===
Generators = 1:9
Consumers = 1:7
Nodes = 1:11
Lines = [
    (1,2),(1,11),(2,3),(2,11),(3,4),(3,9),
    (4,5),(5,6),(5,8),(6,7),(7,8),(7,9),
    (8,9),(9,10),(10,11)
]

# === Mapping of generators and consumers to nodes ===
gen_at_node = [2,2,2,3,4,5,7,9,9]
cons_at_node = [1,4,6,8,9,10,11]

G_node_map = Dict(n => findall(g -> gen_at_node[g] == n, Generators) for n in Nodes)
C_node_map = Dict(n => findall(c -> cons_at_node[c] == n, Consumers) for n in Nodes)

# === Generator and demand parameters ===
unit_cost = Dict(i => [175,100,150,150,300,350,400,300,200][i] for i in Generators)
P_max = Dict(i => [0.02,0.15,0.08,0.07,0.04,0.17,0.17,0.26,0.05][i] for i in Generato
Demand = Dict(j => [0.10,0.19,0.11,0.09,0.21,0.05,0.04][j] for j in Consumers)

# === Line parameters ===
g_line_vals = [4.12,5.67,2.41,2.78,1.98,3.23,1.59,1.71,1.26,1.11,1.32,2.01,4.41,2.14,
b_line_vals = [-20.1,-22.3,-16.8,-17.2,-11.7,-19.4,-10.8,-12.3,-9.2,-13.9,-8.7,-11.3,

g_line = Dict(Lines[i] => g_line_vals[i] for i in 1:length(Lines))
b_line = Dict(Lines[i] => b_line_vals[i] for i in 1:length(Lines))

# === Precompute neighbors for each node ===
neighbors = Dict(n => Int[] for n in Nodes)
for (n1,n2) in Lines
    push!(neighbors[n1], n2)
    push!(neighbors[n2], n1)  # Include reverse link
end

# === Power flow formulas ===
function active_flow(v_k, v_l, theta_k, theta_l, g_kl, b_kl)
    return v_k^2 * g_kl - v_k*v_l*(g_kl*cos(theta_k - theta_l) + b_kl*sin(theta_k - t
end

function reactive_flow(v_k, v_l, theta_k, theta_l, g_kl, b_kl)
    return -v_k^2 * b_kl + v_k*v_l*(b_kl*cos(theta_k - theta_l) - g_kl*sin(theta_k -
```

```
end

# === Optimization Model ===
model = Model(Ipopt.Optimizer)

# === Variables ===
@variable(model, 0 <= Pg[i in Generators] <= P_max[i])
@variable(model, -0.03*P_max[i] <= Qg[i in Generators] <= 0.03*P_max[i])
@variable(model, 0.98 <= V[n in Nodes] <= 1.02)
@variable(model, -pi <= Theta[n in Nodes] <= pi)

# Reference bus
@constraint(model, Theta[1] == 0.0)

# Initial guesses
for n in Nodes
    set_start_value(V[n], 1.0)
    set_start_value(Theta[n], 0.0)
end

# === Objective: minimize active power cost ===
@objective(model, Min, sum(unit_cost[i]*Pg[i] for i in Generators))

# === Power balance constraints ===
for n in Nodes
    g_neigh = [ get(g_line,(n,nb), get(g_line,(nb,n),0.0)) for nb in neighbors[n] ]
    b_neigh = [ get(b_line,(n,nb), get(b_line,(nb,n),0.0)) for nb in neighbors[n] ]

    @NLconstraint(model,
        sum(Pg[i] for i in G_node_map[n]) - sum(Demand[j] for j in C_node_map[n]) ==
        sum(active_flow(V[n], V[neighbors[n][i]], Theta[n], Theta[neighbors[n][i]], g
    )

    @NLconstraint(model,
        sum(Qg[i] for i in G_node_map[n]) ==
        sum(reactive_flow(V[n], V[neighbors[n][i]], Theta[n], Theta[neighbors[n][i]],
    )
end

# === Solve ===
optimize!(model)

# === Results ===
println("\nTotal production cost: ", objective_value(model))

println("\nGenerator outputs:")
for i in Generators
    println("G$i: Pg = ", value(Pg[i]), "  Qg = ", value(Qg[i]))
```

```
end

println("\nNode voltages:")
for n in Nodes
    println("Node $n: V = ", value(V[n]), "   = ", value(Theta[n]))
end
```